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Ronald Reagan and the Space Frontier

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The Reagan Space Legacy

As noted in the first chapter of this study, aerospace historian Andrew Butrica in 2003 suggested that “Ronald Reagan’s two terms as president saw the United States undertake more new and more large space initiatives than any previous administration since that of John Kennedy,” and that the programs and policies initiated during the Reagan administration would “cause the 1980s to be remembered as a major turning point in space history.” He concluded that “Reagan would pass on a lasting legacy to the nation’s space program.” By contrast, during the last year of the Reagan administration, journalist William Broad observed in *The New York Times* that Reagan’s legacy in space was “more reach than grasp,” and that Reagan, while “one of the biggest space enthusiasts to occupy the White House,” would leave behind “a host of costly projects and innovative policies” but “little achievement in the heavens,” with a space program characterized by “muddle and disarray.”¹

The preceding chapters provide detailed evidence with which to evaluate these apparently conflicting claims.² In reality, they are not actually mutually exclusive, and both seem to have some degree of validity. While there may have been few short-term payoffs from the projects and policies adopted by the Reagan administration, as Butrica wrote 15 years after Reagan left the White House positive results from Reagan’s initiatives were beginning to appear. The space station by 2003 was permanently occupied by an astronaut and cosmonaut crew, and on its way to completion and to beginning to be the kind of productive research facility that had been anticipated as Ronald Reagan had approved its development 20 years earlier. The space station partnership with Europe, Canada, and Japan had endured during those 20 years as a visible example of nations working together for a common purpose, and had been amplified as Russia joined the program in the 1990s. Cooperation between the United States and Russia in human spaceflight had become a central feature of the space programs of both countries. Working within the policy and regulatory

framework first set up by the Reagan administration, U.S. firms offering commercial launch services had had modest success, and two new entrepreneurial companies, Space Exploration Technologies (SpaceX) and Blue Origin, were just beginning to attempt to enter the space launch market. Several firms were developing commercial Earth observation businesses, and entrepreneurial firms were beginning to once again explore other opportunities for privately financed space ventures. After its return to flight in September 1988, with a revised context for its operation, the space shuttle had been successfully launched 110 times, mostly carrying out missions for the National Aeronautics and Space Administration (NASA), including deploying, repairing, and servicing the Hubble Space Telescope; launching several planetary probes and other robotic spacecraft; carrying the European Spacelab for in-orbit research; rendezvousing nine times with the Russian space station *Mir*; and beginning space station assembly. Thus, Butrica was somewhat justified in his positive assessment of the Reagan administration space legacy, even if just as his book appeared a second fatal shuttle accident destroyed shuttle orbiter *Columbia*, killing another seven astronauts.

There is also justification for Broad's more skeptical assessment. In the years after Reagan left the White House, there was continual criticism of the lack of a clear statement of purpose to guide the American space program; Reagan had not articulated such a purpose, despite repeated suggestions that he do so. Setting a long-term goal in his 1988 National Space Policy "to expand human presence and activity beyond Earth orbit into the solar system" had been too general to give the hoped-for direction to NASA's human spaceflight effort. In the aftermath of the 2003 *Columbia* accident, the *Columbia* Accident Investigation Board deplored "the lack, over the past three decades, of any national mandate providing NASA a compelling mission requiring human presence in space." The Reagan administration had not succeeded in providing such a mandate.³

The "disarray" with respect to the overall purpose of NASA's efforts that Broad had noted in 1988 had persisted. NASA was frequently accused of "drifting." Ronald Reagan in his space speeches had occasionally spoken of rapid movement into space by the U.S. private sector; that had not happened, as private sector access to space for research purposes was lost after the *Challenger* accident and as the investment community lost its enthusiasm regarding the potential for rapid and large profits from space-based activities. The space station had been repeatedly delayed and subject to frequent redesigns.

Ronald Reagan was a romantic in his attitude toward space achievement, seeing it as a positive part of human progress that was destined to go forward under U.S. leadership. He thus perpetuated in his public remarks an extremely positive view of the potential of space in both tangible and intangible dimensions, and of the central role of human spaceflight in achieving that potential. Those who believed that the focus of space activities should be on exploitation leading to Earth-oriented payoffs, rather than continued human exploration, were critical of Reagan's endorsing the notion that the space program was "the last frontier."⁴

Reagan's enthusiasm for the space program as a frontier activity was a genuine part of his personality and of his ability to lead. However, that enthusiasm was held in check both by Reagan's concern for government fiscal responsibility and by his "bottom up" decision-making approach, which gave space skeptics equal voice with fellow enthusiasts in preparing issues for presidential decision. Only occasionally, as in the case of the space station, did Reagan allow his personal preferences to overrule the counsel of his senior advisers or lead to space budget increases. This reality led to a persistent gap between Reagan's space-related rhetoric and the actual policies and programs of his administration.

ASSESSING THE REAGAN SPACE LEGACY

This study has given particular attention to five areas of space decisions during the Reagan administration. Those decisions were: (1) related to the space shuttle; (2) related to a space station; (3) related to commercial space activities; (4) related to international space cooperation; and (5) related to the overall purpose and goals of the U.S. civilian space program. The Reagan administration legacy in each of these areas is assessed below.

Ultimately, a Realistic Shuttle Policy

During the Reagan administration, more policy attention was given to the space shuttle than to any other element of U.S. space activities. The administration inherited and initially accepted the belief, dating from the origins of the shuttle program, that space shuttle operations would become frequent, routine, and relatively inexpensive. It based its shuttle policy on that belief for several years, in the face of increasing evidence that it was basically a myth. NASA engineers and their industrial contractors had been during the 1970s able to develop a technologically advanced, highly capable shuttle, but not one that could be operated on anything like a frequent, routine, or low-cost basis.

In the early years of the Reagan administration, as the shuttle began flights, the policy framework for operating it was based on the inherited myth rather than the emerging technological reality. This was not totally surprising, since many in the NASA human spaceflight community were deeply vested in that myth. Given prior successes, they had begun to exhibit what Gary Brewer in a perceptive analysis written soon after the *Challenger* accident characterized as "self deception, introversion and a diminished curiosity" about information coming from outside that community. Given the centrality of the space shuttle to NASA's identity and indeed to the objectives of national space policy, it was not in the interest of those responsible for the shuttle to heed warnings that it was a risky system, difficult and expensive to operate safely, and that a policy of sole dependence on the shuttle for U.S. government access to space was deeply flawed. As Brewer observed, "If there is only one game in town, even the hint of failure cannot be tolerated." The message that NASA was carrying out a

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flawed policy was ignored by the NASA leadership. Even as every signal from those outside of NASA who were aware of the shuttle's problems and limitations "shouts 'change,'" suggests Brewer, the "institution plows resolutely along."⁵ This was the case as James Beggs so strongly resisted Pete Aldridge's campaign to provide an expendable complement to the shuttle for national security launches, as advocates of commercializing expendable launch vehicles (ELVs) struggled to remove the shuttle from competing for commercial launches, and as NASA pushed to increase the shuttle flight rate to an unrealistically high level. It took a long time for the Reagan administration to recognize its policy for operating the shuttle was from its inception ill-conceived.

Brewer also observed that "it seems to require the shock of heavy cannon to loosen dissent in a perfect place."⁶ Certainly the tragic loss of seven people in the *Challenger* accident qualifies as "the shock of heavy cannon," and ultimately did lead to a more realistic policy for space shuttle use. Even with the accident, NASA's new administrator, James Fletcher, insisted on the need for replacing *Challenger* with a new shuttle orbiter and resisted as long as possible the multiple pressures to remove the shuttle from its role as a launcher for fee-paying users. Fletcher was likely a particularly poor choice to adapt NASA's shuttle policy to a post-*Challenger* reality, given that he had been the agency's head as the shuttle myth was first perpetuated. He also strongly resented the involvement of other government agencies in what he viewed as NASA's prerogatives, including the guidelines for shuttle operations.

It is an interesting speculation to ask what might have happened had there been no shuttle accident. In principle, Ronald Reagan and his White House associates could have become aware of the shuttle's problems and at some point initiated a change toward a more realistic shuttle policy. Secretary of Defense Cap Weinberger wrote Reagan on January 23, 1984, two years before the *Challenger* accident, saying that the country had made a "serious mistake" in depending on the shuttle for sole access to space. Pete Aldridge waged a vigorous and visible campaign to end national security dependence of the shuttle. Advocates of ELV commercialization were vocal in their criticism of a "shuttle only" policy. But President Reagan had himself become strongly committed to viewing the shuttle's human spaceflight capabilities as an indication of American exceptionalism, and he was not likely easily to change that perspective.

The reality, of course, is that the accident did happen. Even though it took an intense bureaucratic struggle and what many saw as an excessively long time to come up with a realistic post-*Challenger* space transportation policy, the Reagan administration developed and promulgated such a policy. That policy, while employing ELVs for missions not requiring human presence, allowed NASA, once the shuttle returned to flight, to take full advantage of the shuttle's unique capabilities for its missions, and kept the space shuttle in operation for another 20-plus years as a visible symbol of U.S. space leadership. Thus, ultimately developing a realistic policy for space shuttle use must be counted as a positive Reagan space legacy, even if it was preceded by five years of operating the shuttle under false premises and a preventable accident that took seven lives and took the shuttle out of service for almost three years.

Taking the Logical Next Step

In his book *The Space Station*, Hans Mark tells of a December 3, 1983, telephone conversation with James Beggs during which the NASA administrator informed his deputy that President Reagan had approved going ahead with the space station. Mark's response to this message was that "this was a historic occasion" and that "only twice before, in May 1961 for Apollo and in January 1972 for the space shuttle, had a president set a major new direction for the American space program." He added that the decision "spoke well for the political vision and sense of history that President Reagan possesses."⁷ Indeed, the decision to make a permanently occupied space station a centerpiece of American human spaceflight for what has by now been more than three decades was a very significant step.

It was not, however, "a major new direction." NASA in 1970 had deferred, not abandoned, space station development, recognizing that developing a space shuttle first would eventually lead to that development. Creating a human outpost in Earth orbit had been central to NASA's plans from the space agency's very earliest years. What the space station decision represented was, as Beggs was fond of saying, "the logical next step" in a long-existing vision of developing the space frontier. That vision predated the creation of NASA; some even argue that NASA was created to realize it. The vision is centered on a belief that at some point "humans would leave the Earth's surface and explore the universe, just as their ancestors had crossed oceans to explore foreign lands." An orbiting outpost was seen by most space visionaries as a necessary step in the process of human exploration of the solar system, first of all the Earth's Moon and Mars.⁸

James Beggs was a believer in that space vision; he was also an astute political operator. Beggs and Mark had agreed that getting Ronald Reagan to approve a space station would be a major objective during their time at NASA. Beggs had initially hoped that Reagan, with his positive sense of America's future, would quickly embrace the space vision and on that basis early on in his administration would give NASA the go-ahead on the space station. But by the second half of 1982, Beggs made the political judgment that, however personally enthusiastic about the promises of space Reagan might be, other administration officials were very unlikely to recommend that the president approve a space station as a necessary part of achieving an expansive—and expensive—space vision. The NASA leadership and its White House allies, in particular the National Security Council's Gil Rye and Cabinet affairs liaison Craig Fuller, were thus faced with developing a strategy for getting the space station before the president for decision without tying it to accepting a long-term vision for space, even as that vision was actually what made the station "the logical next step."

Their strategy was to make the station decision a test case of the space policy development process adopted early on in the Reagan administration, centered on the Senior Interagency Group for Space [SIG (Space)] as the forum for bringing space decisions to the president. In late 1982, NASA agreed to a year long SIG (Space) study to answer the question, "What policy issues must be identified and resolved in order to establish the basis for an Administration

decision on whether or not to proceed with development of a permanently based, manned space station?" The goal was to have that study completed and a recommendation before the president by November 1983. That schedule was in essence followed, even as strong opposition to the station developed during the year-long study effort.

Beggs and Mark were walking a tightwire in trying to get a positive recommendation on the station out of the SIG (Space) process without tying it to the primary rationale of its being a necessary step toward human exploration beyond Earth orbit. Presidential Science Adviser Jay Keyworth recognized what they were up to, and tried to get NASA to acknowledge the linkage NASA was able to resist, and instead tried to build the case for the station on two main arguments. One was that the space station was a required or facilitating capability for a wide variety of desirable space missions, particularly including ones with potential economic payoffs; the other was that the station was a necessary step beyond the space shuttle to ensure U.S. space leadership in the 1990s and beyond. Lurking in the background was the reality that the Soviet Union had developed a first-generation space station and was on its way to replacing it with a more advanced facility.

The first of these arguments was basically a failure. NASA and its contractors could not come up with evidence that there were a variety of important space missions for which a station was required. Perceptive space veteran Jimmy Hill of the National Reconnaissance Office would comment in mid-1983 "the apparent method used by NASA to develop civil 'requirements' was to list any potential civil space activity (regardless of priority, requirements, cost or technical feasibility) as a civil sector mission for conduct on the space station." In addition, the national security community denied having any need for a station. Thus, the majority of SIG (Space) members were opposed to going ahead with the station program, on the grounds that it was not justified by any set of requirements and that there were better uses for the resources that the station would require.

It was the second, less tangible, "leadership" argument that carried the day with Ronald Reagan. As he made his case for going ahead with the space station to the president on December 1, 1983, Beggs said that a space station would be "a highly visible symbol of U.S. strength" and that "the stakes are enormous: leadership in space for the next 25 years." Reagan resonated to this claim; he in his many space-related speeches had identified U.S. space success with American exceptionalism, and thus was willing to give his approval to the space station even though most of his senior advisers opposed the program. This was unusual behavior for Reagan; on most policy issues he waited until there was some sort of consensus among his inner circle before giving his assent to a course of action. With the respect to the space station, the "leadership" argument indeed convinced Reagan to lead. In announcing his decision to approve station development, Reagan said, "America has always been greatest when we dared to be great. We can reach for greatness again. We can follow our dreams to distant stars, living and working in space."

There were limits to Reagan's support of the space station, however, mostly in terms of the administration's financial commitment to the station program. Another part of the NASA approach to getting Reagan administration approval had been what Howard McCurdy in his book *The Space Station Decision* calls an incremental strategy—NASA would develop a space station “by the yard,” starting with a modest and relatively inexpensive design intended to be supplemented in the future. While the result of NASA's campaign to gain presidential endorsement of the space station was a positive decision, in practice it was not one that got the station program off to a strong start. Ronald Reagan's rhetoric in announcing his decision to approve the station was soaring, but the actual budget for the program in its early years was very modest. McCurdy describes the presidential decision (and that of the Congress in subsequent months) to move forward with the station program as “a very weak commitment,” noting that “policy makers reserved the right to appropriate only enough money to advance the program year by year and to reconsider the purpose of the project as circumstances changed.” This is indeed what happened, as the station went through a series of redesigns during the remaining years of the Reagan administration and there were continuing threats to its budget.⁹

It is valid to question whether the “weak” character of Ronald Reagan's space station decision was appropriate for a program intended to be central to U.S. space activities for several decades. NASA Apollo-era administrator James Webb once described to the author the space station decision as akin to providing an airplane just enough forward momentum to get it off a runway and flying, but continually near stall speed. This was not a recipe for program success. The hesitant character of the space station decision allowed for the instability that plagued the station program during its first decade, until inviting Russia to join the space station partnership gave the program a solid geopolitical rationale.

The space station decision was in some ways similar to that made by the Nixon administration with respect to the space shuttle program. Dale Myers, who had been in charge of the shuttle program at NASA headquarters in the 1970–1973 period as the shuttle program was getting started and who after *Challenger* had returned to NASA as deputy administrator, in 1987 asked Fletcher “whether we are entering the same situation that we entered with the shuttle in 1972.” Then, reflected Myers, “we got starved for budget from the first year on, contributing to design decisions and operational stringencies that might have contributed to the *Challenger* accident.” He and Fletcher wondered whether the station program was headed down a similar path.

While Ronald Reagan's decision to approve space station development did not reflect a commitment of the character required for stability in a long-term undertaking, it *was* a positive decision. Furthermore, when the space station came under congressional attack in 1987 and 1988, Reagan was willing to express his support for the program in communications to key members of Congress. Reagan's rhetoric in support of the space station identified it as central to his vision of the future in space; in January 1987, he declared: “The space

station will be our gateway to the universe, our foothold in outer space, the keystone of our space program. With it as our base camp, we will be able to reach the planets and, perhaps one day, to the stars." It is thus appropriate to identify the space station as Ronald Reagan's most lasting space program legacy.

Laying the Foundation for Space Commercialization

Ronald Reagan and his associates came into office with an ideological conviction that American free enterprise system was a powerful force for ensuring both societal and individual well-being, and that the role of the government was to create a policy and regulatory climate that would allow American business to thrive. This belief intersected with a message coming from entrepreneurs, established businesses, and a NASA eager to justify its activities to the new administration, that there were manifold new business opportunities in space. This conjunction led to the Reagan administration devoting significant policy attention in its first term in office to providing, as stated in its 1982 National Space Policy, "a climate conducive to expanded private sector investment and involvement in civil space activities." That attention was much more driven by ideological belief than it was by either technological or market realities.

Elements of policy attention included White House attempts to privatize government's civilian Earth observation activities; to support the creation of a commercial space launch business based on the use both of the existing ELVs that were to be phased out as the shuttle became the sole launcher for government space missions and of new launch vehicles developed by the emerging entrepreneurial space sector; and to create a series of policy incentives to encourage private sector activity in orbit. On its part, NASA used the potential for high-value activities in orbit as a justification for developing a space station and claimed to be ready to partner with private sector actors in exploring profit-making space activities.

At least a few in the Reagan administration were skeptical of the appropriateness of these commercially oriented initiatives, seeing them, in the words of junior White House staffer Lehman Li, as quite similar to "Democratic proposals for an industrial policy. Government would be making policy targeted toward a specific industry. Industry would be proposing the government policy changes needed to improve the industry's competitiveness." But such skepticism was overridden by enthusiasm for what a former secretary of commerce described as "working factories whose commercial potential has been tested. Properly exploited they will help transform the nation's \$100 billion investment in the space program into sustained world leadership in an area of technology with as yet unimaginable applications for commerce and manufacture." Forecasts of multibillion dollar annual sales of products manufactured in the microgravity environment of space grabbed White House attention.

Beginning in 1981, the White House encouraged efforts to find ways to privatize both meteorological and remote sensing Earth observation activities, ultimately leading to a 1985 transfer of ownership of U.S. Landsat remote

sensing satellites to a private operator. By 1992, that initiative had failed, and the satellites were returned to government ownership and operation. In 1983, President Reagan approved a policy directive intended to facilitate commercialization of ELVs; an Office of Commercial Space Transportation was established in the Department of Transportation to implement that directive. In 1984 Reagan issued another directive setting out the steps government would take to help business move into space. NASA in 1984 created an Office of Commercial Programs; the same year, the 1958 Space Act was amended to direct NASA "to seek and encourage, to the maximum extent possible, the fullest commercial use of space."

There was a fundamental problem with all this activity. It was based on ideology and hope, not reality. Warnings that the market for Earth observation data would not support a private business without government subsidy were ignored in the push toward privatization. While steps to commercialize space launch were taken, they were paralleled by White House approval of marketing the space shuttle as a launcher for fee-paying customers, a step described as a "paradoxical policy that would stymie the creation of a domestic launch industry." There were few demonstrations of the technical feasibility of most of the new space products with purported high economic payoffs, and little, if any, independent analysis of whether those products could compete with Earth-bound equivalents, given the high costs of operating in space. Writing at the 1984 height of the enthusiasm for space commercialization, this author suggested "large, new, commercial payoffs are still 15-20 years away; before then, the cupboard will be almost bare." Even that analysis was optimistic; payoffs from most of the commercial space activities touted in the early 1980s are just beginning to appear, and an industry based on materials processing in the microgravity space environment has never appeared.

Any chance that there would be short-term breakthroughs leading to commercial payoffs were also a casualty of the 1986 *Challenger* accident and the subsequent cutoff of private sector access to space for commercially oriented research. In addition, the date for space station entry into service was by the final years of the Reagan administration rapidly receding. David Thompson, president of the then-new entrepreneurial space company Orbital Sciences, suggested in late 1986 that "we are on the verge of a major policy failure. Our visions of sugar plums have stayed just that." By the end of the Reagan administration, *Aviation Week* was reporting that space commerce had been "stripped of the allure and gold rush claims of the early 1980s" and that the adjusted perception of the long-term and high-risk character of commercial space activities had "slowed or eliminated institutional investment interest" in those activities.¹⁰

While actual economic payoffs from space commercialization are thus not part of the Reagan space legacy, that does not mean that there were no lasting impacts of Reagan administration attention to that issue. Although a commercial launch business was slow to develop, the combination of the 1983 decision to support ELV commercialization and the 1986 decision taking commercial and foreign satellites off of the shuttle after the *Challenger* accident did allow the industry to survive.

More significantly, the early attention to space commercialization created the expectation that there would at some point in time be opportunities for private sector activities in space, and that the government had an important role in facilitating those opportunities. Many of the policy incentives discussed in the 1980s, such as government-industry partnerships and government procurement of space services rather than space hardware, are now in use. The emergence of a dynamic private space sector in recent years can trace its heritage to the Reagan administration encouraging the first-generation space entrepreneurs of the 1980s. The development of a commercial space sector can appropriately be considered a Reagan space legacy.

Another lasting impact of the Reagan administration attention to space commercialization was legitimizing government agencies other than NASA (and the Department of Commerce's National Oceanic and Atmospheric Administration, NOAA) as participants in civilian space policy development. The Department of Commerce and the Department of Transportation have stayed active in promoting and regulating private sector space activities, and continue to participate in interagency space policy processes. In their roles as NASA antagonists during the Reagan administration, representatives of both departments arguably played a productive role in forcing the space agency to begin to adapt to post-*Apollo* reality, a process still underway almost a half century after Americans traveled to the Moon.

Leading Through Cooperation

Reagan administration promotion of international space cooperation is perhaps Ronald Reagan's most clear-cut positive space legacy. Despite almost paranoid concerns among some officials in the Defense and State Departments regarding the possibility of space cooperation leading to unwanted technology transfer, the administration made cooperation in the space station a high-profile activity, advocating it at successive summit gatherings of the leaders of the United States and its closest allies. The international partnership in the space station is today held out as a model of successful cooperation, not only for future space activities, but for other high-technology undertakings. The fact that the partnership existed was an important influence in protecting the space station from cancellation at several points in its history. Persistent administration efforts after 1984 to reestablish space cooperation with the Soviet Union, while producing only modest achievements while Reagan was president, served as the foundation for increasing collaboration with post-Soviet Russia, leading to the 1993 merger of each country's space station program.

By demonstrating U.S. leadership through space cooperation, the space program remained an important instrument of U.S. foreign policy. In the run up to the space station decision, NASA's Beggs noted that "the space station uniquely lends itself to international cooperation. If we can attract that international cooperation, then other nations will be cooperating with us in the resources they spend, rather than competing with us." The Reagan administration recognized this reality, and acted on it.

While there have been a number of tangible benefits to the U.S. space program from cooperation with other spacefaring countries, a primary payoff from that cooperation has been its contribution to the perception of U.S. space leadership in a world of the increasingly capable space efforts of other countries. Assistant Secretary of State John Negroponte in 1987 told the NASA Advisory Council Task Force on International Relations in Space that U.S. space activities "remain flagships of the American scientific and technological enterprise. Our space achievements, and the perception that we have greater capability than any other nation to act in space, have been important elements of our symbolic and world leadership." Space leadership, he stressed, supports the perception that the United States is "the most attractive partner for association and cooperation." This situation persists today.¹¹

The Search for Purpose

President-elect Reagan's space transition team in December 1980 recommended "that the purpose and direction of the U.S. space effort be defined." Eight years later, as a new president prepared to take office, the National Academies of Science and Engineering chartered an ad hoc Committee on Space Policy to answer the question: "What are the U.S. purposes in space?" Creating this committee was an indication that, at least in the judgment of the leaders of the U.S. technical elite, the Reagan administration had failed to follow the advice of its transition team, and that the nation's space program was still without clearly understood "purpose and direction."

In its report, the Committee on Space Policy (this author was a member of the committee) suggested that "a key problem in U.S. civil space policy is the lack of a widely understood purpose, direction, and time scale," in particular for the U.S. human spaceflight effort. The committee also observed that "the lack of long-term, widely-supported, attainable goals ... has led the U.S. space program to promise greater performance than its resources enabled it to deliver."¹²

Was this a fair criticism of the overall Reagan space legacy—leaving behind as it left Washington a space program without a clearly understood purpose or direction, promising more than it could deliver? Certainly, over the eight years that Ronald Reagan was in the White House, there had been repeated calls for setting a dramatic space goal such as a return to the Moon or beginning to prepare for human voyages to Mars. There was intense criticism from space-interested members of Congress and the broader space community, particularly after the *Challenger* accident, of the lack of presidential leadership in defining a future direction for U.S. space efforts. One of the primary complaints regarding the decision to develop a space station was that it, like the space shuttle, was an infrastructure capability, a means to achieve broader goals, not a goal in itself.

There were available for choice well-conceived proposals for the future that the Reagan administration could have embraced. Among the many reviews of the space program that had been carried out during the Reagan administration,

both the Ride report, *Leadership and America's Future in Space*, and the National Commission on Space report, *Pioneering the Space Frontier*, had proposed a "purpose and direction" for the U.S. space effort based on human travel to the Moon and Mars.

The detailed account of space policy development provided in this study suggests that the question of whether to adopt an ambitious space goal beyond the space station was never presented to Ronald Reagan for decision. It is not clear how Reagan would have reacted to such a proposal; given his rhetoric, there is some possibility he might have approved it. Because Reagan seldom took the initiative in proposing policy or program actions (the Strategic Defense Initiative being a notable exception), this implies that there was a continuing judgment among Reagan's advisers, aware of the many proposals for such a goal, that such an initiative was neither in the country's interest nor affordable in the context of the administration's fiscal policies. As the Democrats took control of the Congress, it was also judged not politically feasible. While a lower-level White House official like Gil Rye at the National Security Council may have hoped to get senior-level approval of significant space budget increases as part of creating a National Space Strategy, that hope was from the start unrealistic, given the administration's other priorities and political realities. Science Adviser Jay Keyworth's call for setting out the ambitious goals that a space station was intended to serve also found few supporters.

An additional constraint on adopting an ambitious space goal was the unanticipated multibillion dollar cost of replacing *Challenger*, procuring ELVs for national security and occasional NASA use, and redesigning expensive national security satellites so that they could be launched on those ELVs. The National Academy Committee on Space Policy commented that in the final years of the Reagan administration, "faced with spending constraints for all scientific and technological endeavors, the resources required to correct past policy mistakes—especially sole reliance on the space shuttle for access to space—have been diverted from more forward-looking efforts."¹³

The actual Reagan legacy with respect to articulating a "purpose and direction" for the U.S. space program was far from ambitious. The final Reagan National Space Policy did set out as a "long-range goal" expanding "human presence and activity beyond Earth orbit into the solar system." NASA's Fletcher claimed that "this is a goal of enormous significance with potentially historic future implications." His enthusiasm for the statement was not widely shared; most observers judged the statement to have little practical impact on space program activities. The final Reagan administration budget provided only a very modest \$100 million to begin developing the technology needed for future exploration.

The absence of a defined long-range goal, particularly in terms of identifying one or more destinations for future travel, may well have been politically realistic; it is not at all clear that beyond the space community there would have been public or political support for such goal. Here the experience of Reagan's successor as president, George H.W. Bush, is instructive. On July 20, 1989, the

20th anniversary of the first lunar landing and exactly six months after becoming president, Bush set forth the kind of space vision that advocates had hoped to see from Ronald Reagan. He declared: "We must commit ourselves anew to a sustained program of manned exploration of the solar system and, yes, the permanent settlement of space." Steps toward that goal, he said, would be "first, for the coming decade, for the 1990's: Space Station Freedom, our critical next step in all our space endeavors. And next, for the new century: back to the Moon; back to the future. And this time, back to stay. And then a journey into tomorrow, a journey to another planet: a manned mission to Mars."¹⁴

The Bush proposal became known as the Space Exploration Initiative. Even though the Bush White House, under the leadership of a reestablished National Space Council, struggled to gain traction for the initiative, it was essentially stillborn.¹⁵ This suggests that the Reagan administration's not setting out a clear "purpose and direction" for the civilian space program was not a policy failure, but rather a realistic judgment of the likely fate of such a step. A space effort with ambitions beyond returning the shuttle to regular operation and developing space station *Freedom* was not to be a Reagan leadership legacy.

RONALD REAGAN AND THE SPACE FRONTIER

The notion of space as a "final frontier" was pervasive in Ronald Reagan's presidential rhetoric. This study has included many examples of that rhetoric. To Reagan, the concept of America's role in exploring frontiers was part of his positive, optimistic view of both U.S. history and the country's future.

It must be noted that this view has in recent years become increasingly controversial. Historian Roger Launius, for example, comments that "the construct of the frontier as a positive image of national character and of the progress of democracy has been challenged on all quarters and virtually rejected as a useful ideal in American postmodern, multicultural society." Launius cites another historian, Patricia Limerick, as pointing out that "the frontier myth, used as a happy metaphor by many, should be seen as a pejorative reflection," denoting "conquest of place and peoples, exploitation without environmental concern, wastefulness, political corruption, executive misbehavior, shoddy construction, brutal labor relations, and financial inefficiency."¹⁶

The relevance of the frontier metaphor to depicting space development will not be discussed here. Rather, the point is that it was a central element in the way Ronald Reagan thought about U.S. space achievements, and shaped his positive view of the link between space leadership and American exceptionalism. One of Reagan's biographers notes that "Reagan wasn't acting when he spoke; his rhetorical power rested on his wholehearted belief in all the wonderful things he said about the United States and the American people, about their brave past and their brilliant future." Activities on the space frontier were part of that future.

Another constant theme in Ronald Reagan's space rhetoric was "leadership." That objective may in fact have been, in Reagan's view, a sufficient declaration of purpose to guide the U.S. space effort. As long as the United States

could claim a leading position with respect to other spacefaring countries, Reagan did not appear to push for a more ambitious program. When he was advised that the Soviet Union was threatening to surpass the United States in a particular area of space activity, such as developing a space station, he was willing to approve a U.S. response. One phrase that appeared in many speeches was, "We are first; we are the best; and we are so because we're free."

Even so, it was not leadership at any price. The final Reagan National Space Policy declared that "a fundamental objective guiding United States space activities has been, and continues to be, space leadership. Leadership in an increasingly competitive international environment does not require United States preeminence in all areas and disciplines of space enterprise. It does require United States preeminence in key areas of space activity." There were limits on spending to achieve the leading position in space.

Those limits were evident in NASA's budget allocations. When the Reagan administration took office in January 1981, the NASA budget sat at 0.84 percent of overall government spending; as the administration left the White House in January 1989, it was fundamentally unchanged, at 0.85 percent. This level budget share reflects an underlying reality. As much as Ronald Reagan personally favored the U.S. space program, he and his associates were not willing to increase its priority vis-à-vis other contenders for government support. In this, the Reagan administration was following the dictates of what this author has labeled the "Nixon space doctrine." In his March 1970 statement on the post-*Apollo* U.S. space program, Nixon had said "our space expenditures must take their proper place within a rigorous system of national priorities. What we do in space from here on in must become a normal and regular part of our national life." Ronald Reagan in effect accepted this policy approach as a guide to his space budget decisions; a top Reagan priority was controlling the increase in, if not reducing, government spending.

Many in the space community had hoped, as Ronald Reagan became president, that he would adopt a Kennedy-like attitude toward the space program, being willing to approve bold new initiatives to recapture the spirit of the Apollo program that had dissipated during the 1970s. What they got instead was a president who matched John F. Kennedy in rhetoric but was more similar to Richard Nixon in practice. The Nixon administration had limited the NASA budget to a significantly lower level than at the peak of Apollo, and within that limited budget had approved a major new human spaceflight program, the space shuttle. The Reagan administration kept the NASA budget at the steady, low level where it had settled during the Nixon administration, and within that constrained budget had approved another major human spaceflight program, the space station.

There were, however, important differences between Richard Nixon and Ronald Reagan in their attitude toward the space program. Even as the United States during the time he was in the White House carried out nine flights to the Moon, six of them landing Americans on the lunar surface, Richard Nixon made few space-themed speeches and seldom portrayed the Apollo astronauts

as American heroes. Ronald Reagan continually used the space program as an example of American greatness and consoled the nation when seven astronauts died in the *Challenger* accident. While many internal to the space community criticized Reagan for lack of policy leadership in the post-*Challenger* period, it was his eloquent speech on the evening of the accident that the American public most remembered. Nixon was remembered as the person who ended U.S. space exploration; Reagan, as a president who at the end of his presidency claimed that "in the next century, leadership on Earth will come to the nation that shows the greatest leadership in space."

The Reagan space legacy, then, is a combination of memorable rhetoric and largely pragmatic practice. It is an exaggeration to claim that the eight years of the Reagan administration represented "a major turning point in space history." Rather, it was a period during which the space shuttle went from the centerpiece of access to space to an extremely capable system carrying out cutting edge NASA missions and serving to help construct a permanent outpost in orbit; during which the program to develop that outpost was begun; during which the foundational policies to support private sector space activities were first developed; and during which lasting international space partnerships were forged. Perhaps Ronald Reagan expressed it best as he on January 11, 1989, bid farewell to the country and to his associates in the "Reagan revolution," saying, "We did it. We weren't just marking time. We made a difference ... All in all, not bad, not bad at all."¹⁷

NOTES

1. Andrew J. Butrica, *Single Stage to Orbit: Politics, Technology, and the Quest for Reusable Rocketry* (Baltimore, MD: The Johns Hopkins University Press, 2003), 13. William Broad, "Reagan's Legacy in Space: More Reach Than Grasp," *NYT*, May 8, 1988, IV-9.
2. When material from earlier chapters is quoted in this analysis, the citations to original sources will not be repeated.
3. Columbia Accident Investigation Board, *Report*, Volume I, August 2003, 209.
4. An example of such criticism is in Alex Roland, "Barnstorming in Space: The Rise and Fall of the Romantic Era of Spaceflight, 1957-1986," in Radford Byerly, ed., *Space Policy Reconsidered* (Boulder, CO: Westview Press, 1989).
5. Gary Brewer, "Perfect Places: NASA as an Idealized Institution" in Byerly, *Space Policy Reconsidered*, 159, 163, 166.
6. *Ibid.*, 159.
7. Hans Mark, *The Space Station: A Personal Journey* (Durham, NC: Duke University Press, 1987), 190-191.
8. For a discussion of the influence of the space vision on U.S. space policy, see Howard McCurdy, *Space and the American Imagination* (Washington, DC: Smithsonian Institution Press, 1997). The quote is on 1.
9. Howard McCurdy, *The Space Station Decision: Incremental Politics and Technological Choice* (Baltimore, MD: Johns Hopkins Press, 1990). The quote is on 233.

10. "Washington Roundup," *AWST*, October 6, 1986, 17; "Space Commerce Chastened" and Nicholas Kernstock, "Most Investors Shun High Risk Space Ventures," *AWST*, December 19, 1988, 7, 45.
11. Negroponte is quoted in Task Force on International Relations in Space, NASA Advisory Council, *International Space Policy for the 1990s and Beyond*, October 12, 1987, 20.
12. Committee on Space Policy, National Academy of Sciences and National Academy of Engineering, *Toward a New Era in Space: Realigning Policies to New Realities* (Washington, DC: National Academy Press, 1988), 22, 2, 1.
13. *Ibid.*, 2.
14. George Bush: "Remarks on the 20th Anniversary of the Apollo 11 Moon Landing," July 20, 1989. Online by Gerhard Peters and John T. Woolley, APP, <http://www.presidency.ucsb.edu/ws/?pid=17321>.
15. For an account of the fate of the Space Exploration Initiative, see Thor Hogan, *Mars Wars: The Rise and Fall of the Space Exploration Initiative* (CreateSpace Independent Publishing Platform, 2013).
16. Roger D. Launius, "The Declining Significance of the Frontier in Space History?" July 30, 2012, <https://launiusr.wordpress.com/2012/07/30/the-declining-significance-of-the-frontier-in-space-history/>.
17. Ronald Reagan: "Farewell Address to the Nation," January 11, 1989. Online by Peters and Woolley, APP, <http://www.presidency.ucsb.edu/ws/?pid=29650>.